Lab 6: Rockfish Habitat Modeling

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Introduction

The purpose of this report is to provide the California Department of Fish and Wildlife (CDFW) with a recommendation for a new Marine Protected Area (MPA) in Carmel Bay, California. This recommendation is based on an analysis of bathymetric data that has been previously collected along with Blue Rockfish observations and random observations within the aforementioned study area, as the new MPA will ideally be targeted as one with a reasonable amount of habitat for this species. This data was first analyzed using a Generalized Linear Model (GLM) in R and then plugged into ArcGIS Pro to map overall habitat suitability for the study area. This suitability was then used to determine which of two potential MPA sites had the most suitable habitat area for Blue Rockfish.

Methods

We first needed to sample the environmental variables at known Blue Rockfish locations within the study area. In order to do this, we used the Sample tool in ArcPro to sample rasters based on bathymetry, benthic habitat, distance from shelf break, distance from kelp bed, and benthic complexity. Because of the narrow range of rockfish observations within the entire study area, we then generated “pseudo-absence” point data for different points throughout the rest of the area. These are imperative in determining what kind of habitat is not suitable for rockfish. These points were also sampled for the same five data rasters as the rockfish observations and then placed into a table. The data from these tables was then entered into R to be used in further statistical analysis.

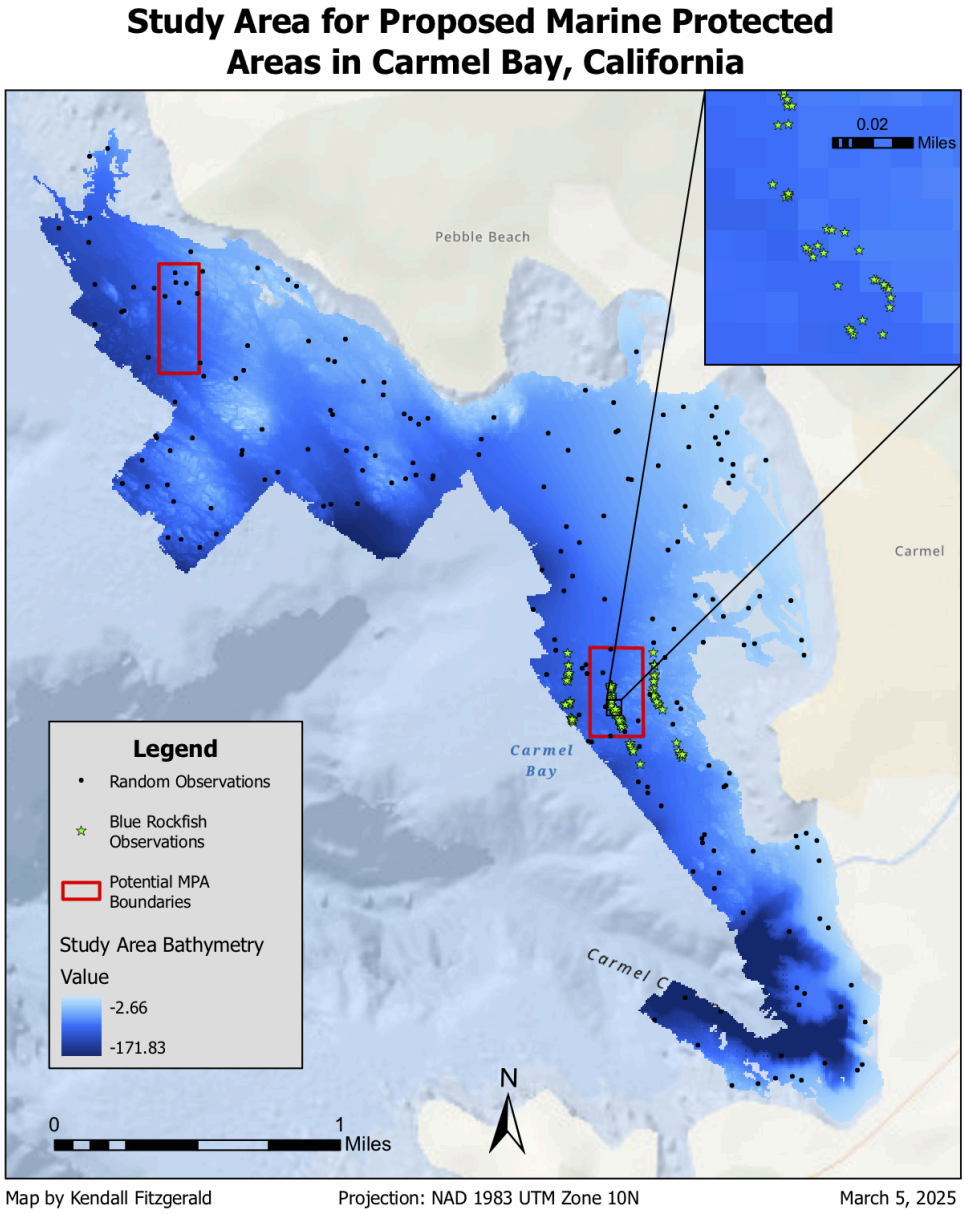
This data was then used to form a Generalized Linear Model, a type of binomial logistic regression. This modeling type was used because we were only looking for two outcomes: suitable habitat for rockfish or not suitable habitat for rockfish. GLM tries to denote the relationship between independent and dependent variables through the formation of a line between these variables. The independent variables in this case are the environmental rasters and the dependent variables are the rockfish observation data points. Because it is a model for the entire study area, there is always a percentage of deviance that expresses how much the model deviates from perfectly fitting the data. The model itself is adjusted through this percentage of deviance along with the Akaike Information Criterion (AIC). The AIC is used to avoid “overfitting” a model so that it can be used in other practical applications. In this case, the AIC analysis determined that removing the distance from shelf breaks as a parameter from the model allowed for best replicability in other capacities.

We then mapped model results back into ArcPro for further geospatial analysis. Using the coefficients for each environmental variable along with the intercept that the model produced, we used the Raster Calculator tool to combine all of these predictors together into a logit prediction

for the entire study area. In order to best interpret the results, we then used the Raster Calculator again to convert the logit prediction into a probability prediction of habitat (with probability ranging from 0 to 100%). Finally, we determined that cells within the study area that met habitat suitability at 50% or over were deemed “Suitable” while any area that had below 50% habitat suitability was “Not Suitable” for rockfish. Our final step involved using the Zonal Statistics tool to determine which of the two possible MPAs had the most suitable habitat for rockfish.

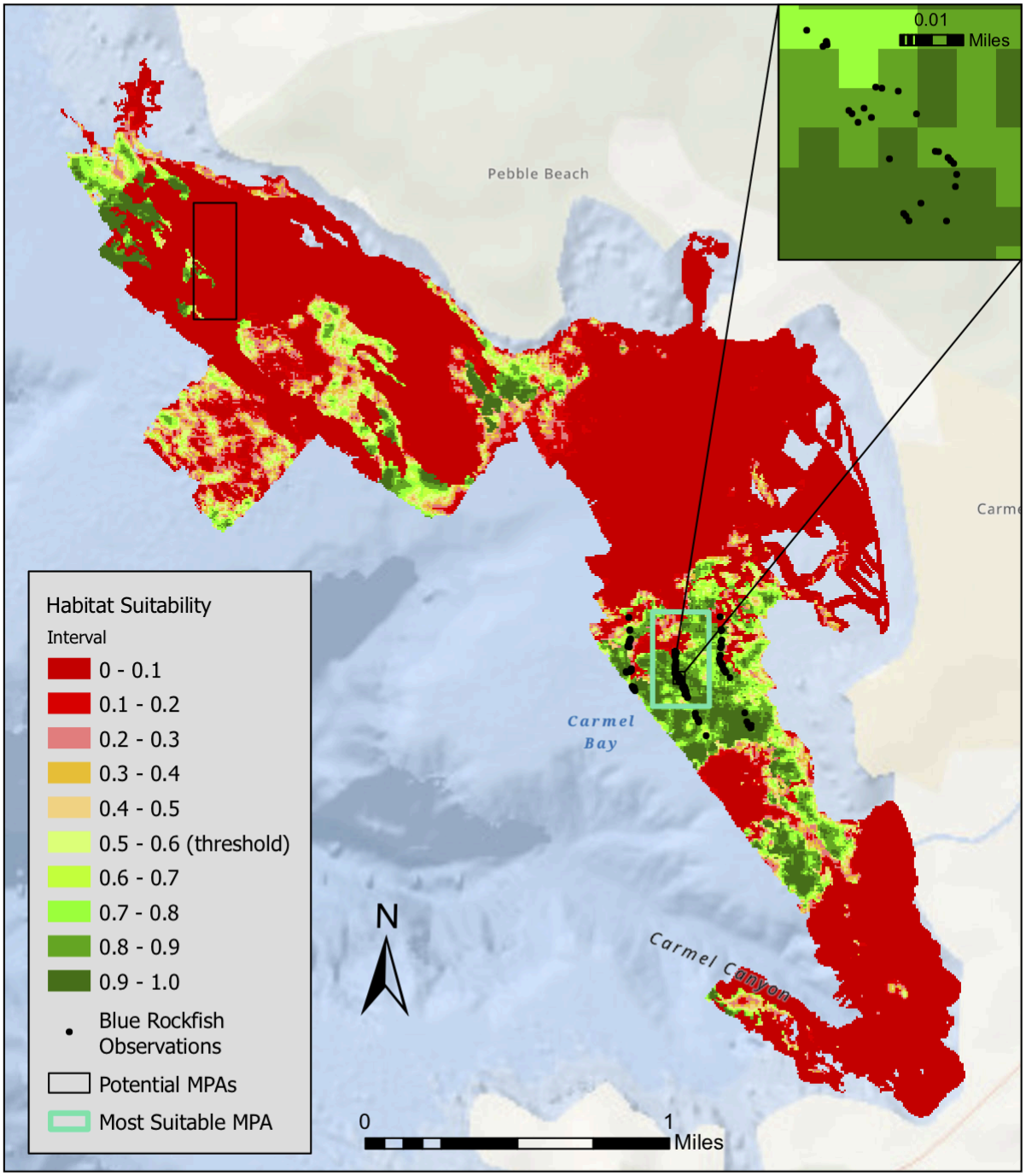
Results and Discussion

Map 1. *The map below is a reference map of the study area in Carmel Bay along with bathymetric data, rockfish and random points data used in the GLM, and the two proposed MPA boundaries.*



Map 2. *The map below shows the probability surface of the GLM model with values ranging from 0 to 1.0 (0 to 100%). Values set at 0.5 (50%) are colored varying shades of green and above show habitat that meets minimum suitability requirements for Blue Rockfish while values below that threshold are colored in varying shades of red to indicate that they are not suitable.*

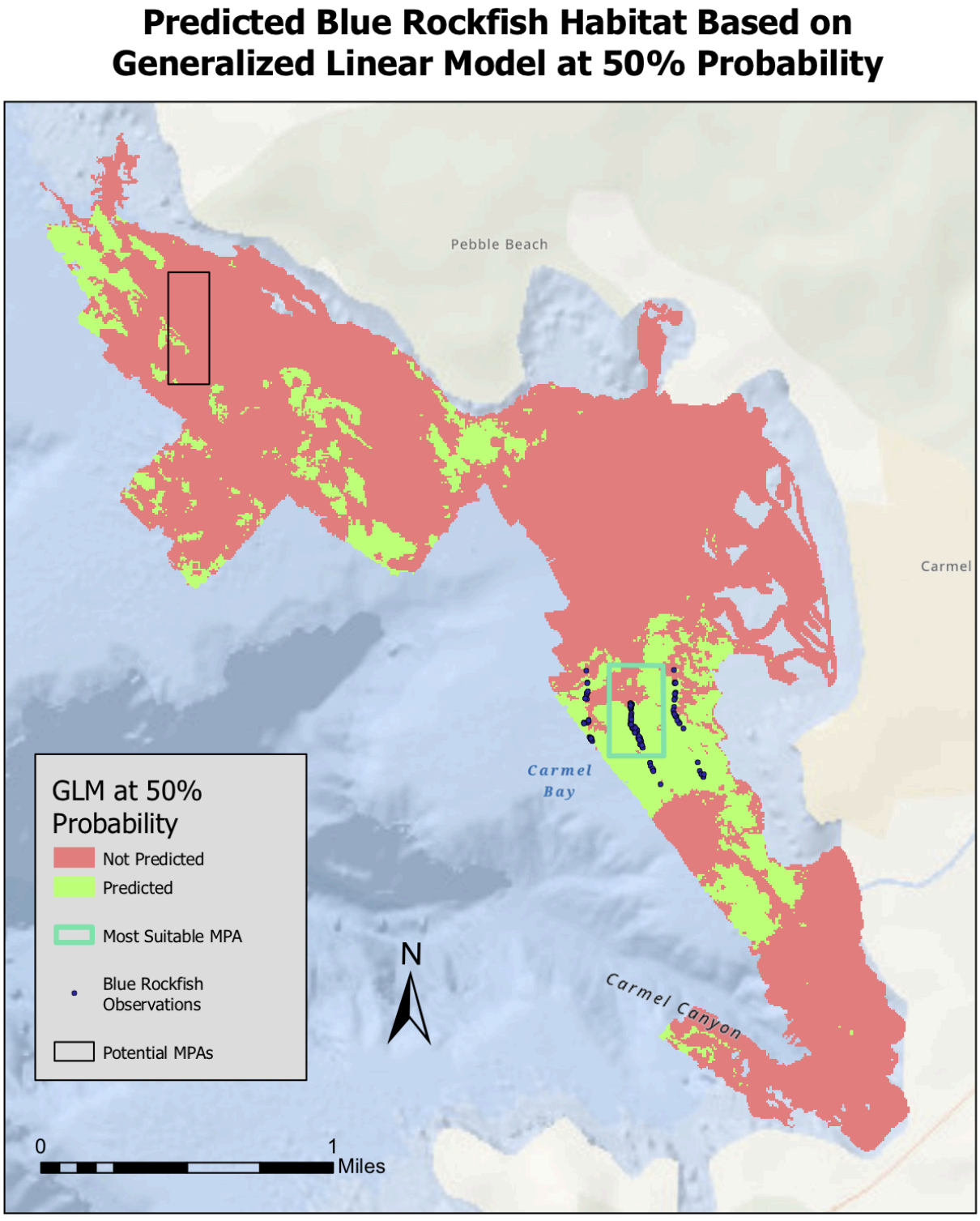
Blue Rockfish Habitat Suitability Based on a Generalized Linear Model of the Study Area



Map by Kendall Fitzgerald Projection: NAD 1983 UTM Zone 10N March 5, 2025

Esri, Garmin, NaturalVue, Sources: Esri, TomTom, Garmin, GEBCO, National Geographic, NOAA, and the GIS User Community

Map 3. *This map shows the final habitat suitability results based on the 50% probability threshold. Areas outlined in red lie below that criteria and are not suitable while areas colored green fall within that threshold and above and are suitable habitat for rockfish. The most suitable MPA that has the most suitable area for rockfish is highlighted in teal.*



MPA Habitat Suitability for Blue Rockfish Based on Generalized Liner Model			
MPA ID	MPA Area (m^2)	Habitat Area in MPA (m^2)	% of MPA in Threshold
1	147,200	112,300	76%
2	134,200	4,200	3%

Table 1. *The table above showcases the final results of the Zonal Statistics tool being run on the two potential MPA sites. The 50% probability threshold for habitat suitability was utilized for this tool and found that MPA 1 was the Most Suitable with 76% of its habitat deemed "Suitable" for Blue Rockfish.*

Map 1 is a simple reference map of the study area. It has a bathymetric layer that shows the bottom complexity in Carmel Bay as well as the rockfish observations and the random points used in generating the "pseudo-absence" observations. In addition, it has the two potential MPA sites that are ultimately being analyzed throughout this report. Map 2 shows the prediction results of the GLM for the study area. Because this prediction spans from 0 to 100% in terms of habitat suitability and the threshold that was set for this analysis was at 50%, the symbology shows this difference accordingly. Areas ranging from 0 to below 0.5 had varying colors of red to indicate that they did not meet the minimum requirements to be suitable for rockfish while areas 0.5 and above showed varying ranges of green to illustrate that they did meet the minimum habitat suitability requirements. The darker red signifies the most unsuitable areas while the darker green signifies the most suitable areas for this species within the study area.

Map 3 shows the final product of deeming areas either "Suitable" or "Not Suitable". Because this result is either one or the other, there is only a 2-color symbology with suitable areas represented in green and unsuitable areas represented in red. This map shows a simplified version of the results while Map 2 shows a more complex range of where the most and least suitable rockfish habitat lies within the study area. Table 1 shows the resulting statistics about the percentage of each potential MPA that has "Suitable Habitat" (over the 50% threshold) within its boundaries. MPA 1 is highlighted in green since 76% of its area has been deemed to have "Suitable" rockfish habitat based on the set parameters and MPA 2 is highlighted in red because only 3% of its area meets the suitability criteria.

Based on our GLM model results we can see that all four environmental predictors that were left in the model (bathymetry, benthic habitat, distance from kelp bed, and benthic complexity) showed a strong correlation between the dependent variable (the presence of rockfish). Within this correlation data, we see that deeper depths result in lower rockfish presence, higher benthic complexity results in higher incidences of rockfish, and smaller distances from kelp beds show higher rockfish presence because it's essential to their survival. It is important to keep in mind that the overall results of this analysis are a model and do not reflect real-time data surveys of an entire area. Because we had limited rockfish observations in a

concentrated area and generated random “pseudo-absence” points about the locations that they weren’t, we should not assume that Blue Rockfish do not exist in environments outside of the green areas in Maps 2 and 3. Because they have the ability to swim around, they can venture outside of boundaries that are determined to be their natural habitat based on data and predictions and further research should be done in this area with more observation points.

That being said, based on the data that was provided and the GLM that was generated, my final determination for the most suitable MPA based on rockfish habitat is MPA 1 (highlighted in teal in Maps 2 and 3). I look forward to working with CDFW to effectively communicate these results to policymakers so that we can move forward with this designation as efficiently as possible.